Lecture 7. Properties of Expectation

Linearity, covariance, conditional expectation, prediction, moment generating functions, multivariate normal

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7.1 Expectation of sums

For jointly distributed X, Y, $E[g(X,Y)] = \sum_x \sum_y g(x,y)p(x,y)$ or $\iint g(x,y)f(x,y)\,dx\,dy$ (proved as Proposition 5.2). Taking $g = x + y$ gives linearity, $E[X+Y] = E[X] + E[Y]$, and by induction $E[\sum X_i] = \sum E[X_i]$, with no independence needed. Also $X \geq Y$ implies $E[X] \geq E[Y]$.

Example 7.1

(a) An accident at $X \sim U(0,L)$ and an ambulance at independent $Y \sim U(0,L)$: $E|X-Y| = \frac{1}{L^2}\int_0^L \int_0^L |x-y|\,dx\,dy = \frac{L}{3}$.
(b) Boole's inequality: if X counts how many of $A_1, \ldots, A_n$ occur and $Y = \mathbb{1}\{X \geq 1\}$, then $X \geq Y$, so $P(\bigcup A_i) = E[Y] \leq E[X] = \sum P(A_i)$.
(c) The sample mean $\bar{X} = \frac{1}{n}\sum X_i$ of identically distributed variables has $E[\bar{X}] = \mu$.

Example 7.2 (Indicator calculations)

(a) Matches: $E[\#\text{matches}] = N \cdot \frac{1}{N} = 1$ for every N.
(b) Coupon collecting with N types: once i types are held, the number of further purchases until a new type is geometric with success probability $(N-i)/N$. So the total T has

$$E[T] = \sum_{i=0}^{N-1} \frac{N}{N-i} = N\Big(1 + \frac{1}{2} + \cdots + \frac{1}{N}\Big) \approx N \log N.$$

The expected number of distinct types in n purchases is $N\{1-(1-1/N)^n\}$, by summing indicators of "type j appears".
(c) Runs: in a random arrangement of n ones and m zeros, the expected number of runs of ones is $\frac{n(m+1)}{n+m}$. Let I_i be the indicator that a run of ones starts at position i; $P(I_1 = 1) = \frac{n}{n+m}$, and for $i > 1$, $P(I_i = 1) = P(\text{position } i-1 \text{ is } 0,\ i \text{ is } 1) = \frac{mn}{(n+m)(n+m-1)}$.
(d) Quicksort sorts n distinct values by choosing a random pivot, comparing all others with it, and recursing on both sides. The i-th and j-th smallest values ($i < j$) are compared iff one of them is the first pivot chosen among the $j-i+1$ values between them inclusive, which has probability $2/(j-i+1)$. Hence

$$E[\#\text{comparisons}] = \sum_{i<j} \frac{2}{j-i+1} = 2(n+1)\sum_{k=1}^{n} \frac{1}{k} - 4n \approx 2n \log n.$$

Proposition 7.1 (Moments of the number of events)

If X is the number of the events $A_1, \ldots, A_n$ that occur, then

$$E\Big[\binom{X}{k}\Big] = \sum_{i_1 < \cdots < i_k} P(A_{i_1} \cdots A_{i_k}).$$

Proof. $\binom{X}{k}$ is the number of k-subsets of the events that all occur, i.e. $\sum_{i_1<\cdots<i_k} I_{i_1} \cdots I_{i_k}$; take expectations. ∎

With $k = 2$, $E[X^2] = E[X] + 2\sum_{i<j} P(A_i A_j)$. For matches, $P(A_i A_j) = \frac{1}{N(N-1)}$, so $E[X(X-1)] = 1$ and $\mathrm{Var}(X) = 1$, as for Poisson(1).

7.2 Covariance and variance of sums

Motivation. The variance of a sum depends on how the terms move together. Covariance captures this, and expanding $\mathrm{Var}(\sum X_i)$ in covariances solves many problems that look hard, such as matching and sampling without replacement.

Proposition 7.2

If X, Y are independent, $E[g(X)h(Y)] = E[g(X)]E[h(Y)]$.

Proof. $\iint g(x)h(y)f_X(x)f_Y(y)\,dx\,dy$ factors. ∎

$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)] = E[XY] - E[X]E[Y]$. Independence implies $\mathrm{Cov} = 0$, but not conversely: for X uniform on $\{-1,0,1\}$ and $Y = X^2$, $\mathrm{Cov}(X,Y) = E[X^3] = 0$.

Proposition 7.3

Covariance is symmetric and bilinear: $\mathrm{Cov}(\sum_i a_i X_i, \sum_j b_j Y_j) = \sum_{i,j} a_i b_j \mathrm{Cov}(X_i, Y_j)$, and $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$. Hence

$$\mathrm{Var}\Big(\sum_i X_i\Big) = \sum_i \mathrm{Var}(X_i) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j),$$

which is $\sum \mathrm{Var}(X_i)$ for pairwise independent variables.

Example 7.3

(a) For i.i.d. X_i with variance σ^2, $\mathrm{Var}(\bar{X}) = \sigma^2/n$, and since $\sum(X_i - \bar{X})^2 = \sum(X_i-\mu)^2 - n(\bar{X}-\mu)^2$,

$$E[S^2] = \frac{1}{n-1}(n\sigma^2 - n\cdot\sigma^2/n) = \sigma^2, \qquad S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2.$$

(b) Hypergeometric variance: with X_i the indicator that the i-th draw is white, $\mathrm{Cov}(X_i, X_j) = \frac{m(m-1)}{N(N-1)} - \frac{m^2}{N^2} = -\frac{m(N-m)}{N^2(N-1)}$ for $i \neq j$, which with n draws gives $\mathrm{Var}(X) = n\frac{m}{N}(1-\frac{m}{N})\frac{N-n}{N-1}$.
(c) Multinomial: if N_i is the number of trials with outcome i, then $\mathrm{Cov}(N_i, N_j) = -np_ip_j$ for $i \neq j$: write $N_i = \sum_k I_i(k)$; $I_i(k)I_j(k) = 0$ and trials are independent, so $\mathrm{Cov}(N_i,N_j) = \sum_k \mathrm{Cov}(I_i(k), I_j(k)) = n(0 - p_ip_j)$.

The **correlation** is $\rho(X,Y) = \mathrm{Cov}(X,Y)/\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}$.

Proposition 7.4

$-1 \le \rho(X,Y) \le 1$, with $\rho = \pm 1$ iff $Y = a + bX$ with probability 1 for some b of the same sign.

Proof. With σ_X^2, σ_Y^2 the variances, $0 \le \mathrm{Var}(X/\sigma_X \pm Y/\sigma_Y) = 2(1 \pm \rho)$. Equality means $X/\sigma_X \pm Y/\sigma_Y$ is constant. ∎

ρ measures *linear* dependence only. For the bivariate normal of Example 6.11, $\rho(X,Y) = \rho$.

7.3 Conditional expectation

Motivation. $E[X \mid Y]$ is the best guess of X once Y is known, itself a random variable. Averaging it gives back $E[X]$, a "divide and conquer" tool for computing expectations by conditioning on a first step.

$E[X \mid Y = y] = \sum_x x p_{X|Y}(x \mid y)$ or $\int x f_{X|Y}(x \mid y)\,dx$. $E[X \mid Y]$ denotes the random variable $g(Y)$ where $g(y) = E[X \mid Y = y]$.

Theorem 7.5 (Tower property)

$E[X] = E[E[X \mid Y]]$, i.e. $E[X] = \sum_y E[X \mid Y = y]P(Y = y)$ or $\int E[X \mid Y = y] f_Y(y)\,dy$.

Proof. Discrete case: $\sum_y \sum_x x p_{X|Y}(x \mid y) p_Y(y) = \sum_x \sum_y x p(x, y) = E[X]$. ■

Example 7.4 (The trapped miner)

A miner chooses at random among three doors: door 1 leads to safety in 3 hours, doors 2 and 3 lead back to the mine after 5 and 7 hours. With X the time to safety and Y the first door, $E[X] = \frac{1}{3}\{3 + (5 + E[X]) + (7 + E[X])\}$, so $E[X] = 15$.

Example 7.5 (Random sums)

If N is a nonnegative integer variable independent of i.i.d. $X_1, X_2, \ldots$, then $E[\sum_{i=1}^N X_i \mid N = n] = nE[X]$, so

$$E\Big[\sum_{i=1}^{N} X_i\Big] = E[N]E[X].$$

(The expected spending in a store is the expected number of customers times the mean spending.)

Example 7.6 (Waiting for k consecutive successes)

In independent trials with success probability p, let N_k be the number of trials until k consecutive successes. Conditioning on N_{k-1}: one more trial either succeeds (done) or fails (start over), so $E[N_k \mid N_{k-1}] = N_{k-1} + 1 + (1 - p)E[N_k]$, giving $E[N_k] = \frac{1}{p} + \frac{E[N_{k-1}]}{p}$ and

$$E[N_k] = \frac{1}{p} + \frac{1}{p^2} + \cdots + \frac{1}{p^k}.$$

For a fair coin, $E[N_2] = 6$ and $E[N_3] = 14$.

Probabilities by conditioning. With $X = I_E$, the tower property reads $P(E) = E[P(E \mid Y)]$. For instance, for independent continuous X, Y, $P(X < Y) = \int F_X(y) f_Y(y)\,dy$, and $P(X + Y < a) = \int F_X(a - y) f_Y(y)\,dy$ is the convolution formula again.

Example 7.7 (The best-prize problem)

n prizes of distinct values appear in random order; each must be accepted or rejected on the spot, knowing only the relative ranks so far. Strategy: reject the first k, then accept the first prize better than all earlier ones. Let X be the position of the best prize. If $X = i > k$, we

win iff the best of the first $i-1$ prizes is among the first k, which has probability $k/(i-1)$. So

$$P_k(\text{best}) = \frac{1}{n}\sum_{i=k+1}^{n}\frac{k}{i-1} \approx \frac{k}{n}\log\frac{n}{k}.$$

Maximizing $x\log(1/x)$ over $x = k/n$ gives $k \approx n/e$, and the success probability tends to $1/e \approx 0.368$ however large n is.

Example 7.8 (Uniform sums revisited)

Let $m(x) = E[N(x)]$, where $N(x) = \min\{n : U_1 + \cdots + U_n > x\}$ for $0 \le x \le 1$. Conditioning on $U_1 = y$: if $y > x$ then $N = 1$, otherwise $N = 1 + N'(x-y)$. So $m(x) = 1 + \int_0^x m(x-y)\,dy = 1 + \int_0^x m(u)\,du$, whence $m' = m$, $m(0) = 1$, and $m(x) = e^x$. At $x = 1$ this recovers Example 6.8.

Proposition 7.6 (Conditional variance formula)

With $\operatorname{Var}(X \mid Y) = E[(X - E[X \mid Y])^2 \mid Y]$,

$$\operatorname{Var}(X) = E\big[\operatorname{Var}(X \mid Y)\big] + \operatorname{Var}\big(E[X \mid Y]\big).$$

Proof. $E[\operatorname{Var}(X \mid Y)] = E[X^2] - E[(E[X \mid Y])^2]$ and $\operatorname{Var}(E[X \mid Y]) = E[(E[X \mid Y])^2] - (E[X])^2$; add. ∎

Example 7.9

(a) For the random sum $S = \sum_1^N X_i$: $E[S \mid N] = N\mu$ and $\operatorname{Var}(S \mid N) = N\sigma^2$, so $\operatorname{Var}(S) = E[N]\sigma^2 + \mu^2\operatorname{Var}(N)$. With $N \sim \text{Poisson}(\lambda)$, $\operatorname{Var}(S) = \lambda E[X^2]$.
(b) Geometric variance: if Y indicates success on the first trial, $E[N \mid Y]$ is 1 or $1 + 1/p$ and $\operatorname{Var}(N \mid Y)$ is 0 or $\operatorname{Var}(N)$; solving gives $\operatorname{Var}(N) = (1-p)/p^2$.

7.4 Conditional expectation and prediction

Motivation. Among all functions of Y, $E[X \mid Y]$ minimizes mean squared prediction error. When we restrict to linear functions, we obtain the regression line.

Proposition 7.7

$E[(Y - g(X))^2] \ge E[(Y - E[Y \mid X])^2]$ for every function g: $E[Y \mid X]$ is the best predictor of Y from X in mean square error.

Proof. Given X, $E[(Y-c)^2 \mid X]$ is minimized by $c = E[Y \mid X]$ (variance $\le$ any second moment about c); take expectations. ∎

If we restrict to linear predictors $a + bX$, minimizing $E[(Y - a - bX)^2]$ gives

$$b = \rho\frac{\sigma_Y}{\sigma_X}, \qquad a = \mu_Y - b\mu_X, \qquad \text{MSE} = \sigma_Y^2(1-\rho^2).$$

For the bivariate normal $E[Y \mid X]$ is linear, so the best linear predictor is the best predictor. Galton's observation that sons of tall fathers are on average less tall ("regression to the mean") is the factor $|\rho| < 1$.

7.5 Moment generating functions

Motivation. The mgf turns sums of independent variables into products and determines the distribution. This makes it the main tool for identifying the distribution of a sum and for proving the central limit theorem.

The **moment generating function** of X is $M(t) = E[e^{tX}]$. When finite near 0, $M^{(n)}(0) = E[X^n]$ (differentiate under the expectation).

Distribution	$M(t)$	valid for
$\mathrm{Bin}(n,p)$	$(pe^t + 1-p)^n$	all t
$\mathrm{Poisson}(\lambda)$	$\exp\{\lambda(e^t-1)\}$	all t
$\mathrm{Geometric}(p)$	$pe^t/\{1-(1-p)e^t\}$	$(1-p)e^t < 1$
$\mathrm{Exp}(\lambda)$	$\lambda/(\lambda - t)$	$t < \lambda$
$\mathrm{Gamma}(\alpha,\lambda)$	$\{\lambda/(\lambda-t)\}^\alpha$	$t < \lambda$
$N(\mu,\sigma^2)$	$\exp\{\mu t + \sigma^2 t^2/2\}$	all t

For the normal: $E[e^{tZ}] = \frac{1}{\sqrt{2\pi}} \int e^{-(x-t)^2/2 + t^2/2}\,dx = e^{t^2/2}$, and $X = \mu + \sigma Z$.

Theorem 7.8

(a) If X, Y are independent, $M_{X+Y}(t) = M_X(t) M_Y(t)$.
(b) (Uniqueness) If $M_X = M_Y$ on a neighbourhood of 0, then X and Y have the same distribution.

(a) is Proposition 7.2 with $g = e^{tx}$, $h = e^{ty}$;

(b) is quoted without proof. Together they make Proposition 6.2 immediate: e.g. $e^{\mu_1 t + \sigma_1^2 t^2/2} e^{\mu_2 t + \sigma_2^2 t^2/2}$ is the normal mgf with parameters $\mu_1 + \mu_2$, $\sigma_1^2 + \sigma_2^2$.

Example 7.10

(a) χ^2_n: $E[e^{tZ^2}] = (1-2t)^{-1/2}$, so $Z_1^2 + \cdots + Z_n^2$ has mgf $(1-2t)^{-n/2}$, the $\mathrm{Gamma}(n/2, 1/2)$ mgf.
(b) Random sums: $E[e^{tS} \mid N] = M_X(t)^N$, so $M_S(t) = E[M_X(t)^N]$. Differentiating at 0 recovers $E[S] = E[N]\mu$ and the variance formula of Example 7.9.
(c) If $M(t) = e^{3(e^t - 1)}$, then $X \sim \mathrm{Poisson}(3)$ and $P(X = 0) = e^{-3}$.

The **joint mgf** $M(t_1, \ldots, t_n) = E[e^{t_1 X_1 + \cdots + t_n X_n}]$ determines the joint law, and $X_1, \ldots, X_n$ are independent iff it factors into the product of the marginal mgfs.

Example 7.11

Let X, Y be i.i.d. $N(\mu, \sigma^2)$. Then $X+Y$ and $X-Y$ are independent: $t(X+Y) + s(X-Y) = (t+s)X + (t-s)Y$, so

$$E[e^{t(X+Y)+s(X-Y)}] = e^{2\mu t + \sigma^2(t^2+s^2)} = E[e^{t(X+Y)}]\,E[e^{s(X-Y)}].$$

7.6 The multivariate normal distribution

If $Z_1, \dots, Z_n$ are independent $N(0,1)$ and $X_i = \mu_i + \sum_j a_{ij} Z_j$ $(i = 1, \dots, m)$, then $(X_1, \dots, X_m)$ is **multivariate normal**. Every linear combination $\sum t_i X_i$ is then normal, with mean $\sum t_i \mu_i$ and variance $\sum_{i,j} t_i t_j \operatorname{Cov}(X_i, X_j)$, so

$$M(t_1, \dots, t_m) = \exp\Big\{ \sum_i t_i \mu_i + \frac{1}{2} \sum_{i,j} t_i t_j \operatorname{Cov}(X_i, X_j) \Big\} :$$

the joint law is determined by the means and covariances. In particular, uncorrelated components of a multivariate normal vector are independent (the mgf factors).

Theorem 7.9 (Sample mean and variance)

If $X_1, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$, then $\bar{X}$ and S^2 are independent, $\bar{X} \sim N(\mu, \sigma^2/n)$, and $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$.

Proof. $(\bar{X}, X_1 - \bar{X}, \dots, X_n - \bar{X})$ is multivariate normal, and $\operatorname{Cov}(\bar{X}, X_i - \bar{X}) = \sigma^2/n - \sigma^2/n = 0$, so $\bar{X}$ is independent of the deviations and hence of S^2. Next,

$$\sum_{i=1}^{n} \Big(\frac{X_i - \mu}{\sigma} \Big)^2 = \frac{(n-1)S^2}{\sigma^2} + \Big(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \Big)^2 .$$

The left side is χ^2_n and the last term is χ^2_1 independent of the first; dividing mgfs, $(n-1)S^2/\sigma^2$ has mgf $(1-2t)^{-n/2}/(1-2t)^{-1/2} = (1-2t)^{-(n-1)/2}$. ∎

7.7 General definition of expectation

For a nonnegative X that is neither discrete nor continuous one defines $E[X]$ as the limit of $E[X_n]$ for simple $X_n \uparrow X$ (the Lebesgue integral), or as $\int_0^\infty P(X > x)\, dx$; for general X, $E[X] = E[X^+] - E[X^-]$. All properties above continue to hold. If $X = Y$ with probability p and W otherwise, $E[X] = pE[Y] + (1-p)E[W]$.